Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

- Introduction to Model Selection
  Reading • 5 min
- Bayesian Model Selection
  Reading • 30 min
- Bayesian Model Selection Readings
  Reading • 1h

Lesson 2: Cross Validation

- Introduction to Cross-Validation
  Reading • 25 min
- K-Fold Cross-Validation
  Reading • 2 min
- Leave-One-Out Cross-Validation (LOOCV)
  Reading • 10 min

Lesson 3: Introduction to Optimization Techniques

- **Introduction to Optimization Techniques**
  Reading • 13 min
- Optimization Using Gradient Descent
  Reading • 40 min
- Gradient Descent with Momentum
  Reading • 45 min
- Stochastic Gradient Descent (SGD)
  Reading • 40 min
- Constrained Optimization
  Reading • 16 min
- Lagrange Multipliers
  Reading • 8 min
- Convex Optimization
  Reading • 8 min
- Linear Programming
  Reading • 15 min
- Quadratic Programming
  Reading • 10 min
- Check Your Knowledge: Optimization Techniques
  Graded Assignment • 15 min

Introduction to Optimization Techniques

0:06 / 1:25 2x

Optimization techniques are fundamental in machine learning and statistical modeling, as they provide the means to find the best parameters or solutions for a given problem. In the context of machine learning, optimization is often about minimizing an objective function, which typically represents the error or loss of a model on a given dataset. The goal is to find the set of parameters that lead to the best possible performance of the model on unseen data. Finding the best value is like finding the valleys of the objective function, and the gradients point us uphill. The idea is to move downhill and find the deepest point.

Optimization Types - Neos Guide (8 minutes)

This resource goes into optimization types in great detail using a flowchart to explain the various solutions that can be used.

Types of Optimization Problems

- Convex Optimization: These are problems where the objective function is convex, meaning any local minimum is also a global minimum. Convex optimization problems are generally easier to solve because they guarantee that any solution found is the best possible solution.
- Non-Convex Optimization: In non-convex optimization problems, the objective function may have multiple local minima and maxima, making it more challenging to find the global minimum. These types of problems are common in complex models like deep neural networks.

Go to next item ✓ Completed

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